

Emon Roy

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Professional Summary

AI-focused Computer Science graduate student with over 3 years of industry experience and 2 years of academic research, specializing in building and evaluating LLM-powered systems. Hands-on experience in prompt engineering, RAG pipelines, and structured evaluation of model outputs for quality and reliability. Experienced running experiments to measure engagement and performance, and collaborating in Git-based teams within fast-paced, cross-functional environments. Seeking a Machine Learning Engineer role to build consumer-facing GenAI features and strengthen model reliability.

Experience

Ontario Tech University

Sep 2023 – Dec 2025

Graduate Research Assistant

Oshawa, Ontario

- Conducted independent research on LLM-powered systems and CNN interpretability under faculty supervision, from literature review through experimental design and manuscript preparation.
- Co-authored two peer-reviewed papers accepted at COMPSAC 2025 and CVIS 2024, managing the submission and revision process alongside the faculty advisor.
- Designed and ran controlled experiments to evaluate model performance and reliability, documenting methodology and results for reproducibility.
- Collaborated with faculty and lab members in a Git-based workflow to manage codebases, track experiments, and review each other's work.

IFIC Bank PLC

Dec 2020 – Aug 2023

Software Engineer

Dhaka, Bangladesh

- Successfully delivered an E-Signature solution for the bank's HR Division that allowed staff members to easily collect and upload client signatures resulting in a significant reduction in the bank's admin approval turnaround time by 60%.
- Spearheaded the development of the bank's branch affairs repository system across its branches that allowed its members to quickly access and report on all end of day bank affairs including summary transactions.
- Administered and supported Oracle databases for banking applications using SQL/PL-SQL, resolving production incidents and completing data validation checks that reduced recurring data issues by 30%.
- Assisted with performance tuning and query optimization using TOAD, reducing report query runtime by 40% and improving operational reporting reliability by 25%.
- Automated HRIS reporting by delivering 10+ BI dashboards, resulting in a 35% increase in operational efficiency for analytics stakeholders.
- Led the research and development on different banking applications and provide on-time technical support and feedback to the branch & sub-branch employees at request.

Ontario Tech University

Sep 2023 – Apr 2026

Graduate Teaching Assistant

Oshawa, Ontario

- Delivered engaging lectures on Scientific Data Analysis, Software Systems Development & Integration, Introduction to Programming, Algorithm Analysis & Design, and Software Quality Assurance to classes of 50+ students, boosting midterm exam averages from 75% to 86%.
- Designed and led interactive tutorials and laboratory sessions, increasing first-attempt pass rates on algorithmic problem sets from 62% to 84% through quick-quiz checkpoints and hands-on examples.
- Proctored exams and provided timely, accurate grading and constructive feedback, driving course-evaluation ratings up from 4.0 to 4.6 out of 5, with 92% of students citing "clarity of instruction" as a strength.

Technical Skills

Programming Languages: Python, C/C++, Java, SQL

AI/LLM Expertise : Hugging Face, OpenAI API, Fine-tuning (GPT, BERT)

GenAI / RAG : Prompt engineering, RAG pipelines

ML/DL Frameworks : PyTorch, TensorFlow, Keras, TorchVision

Libraries & Data Tools : Pandas, NumPy, PySpark, scikit-learn, Matplotlib, Plotly

Evaluation : A/B testing, Offline evaluation, Quality & trust metrics analysis, Failure mode analysis

Deployment : AWS/Azure, FastAPI, Docker, Git, CI/CD

Projects

Transformation-Invariant Grad-CAM in CNNs [\[Github\]](#) | Python, TensorFlow/Keras, OpenCV, scikit-learn, NumPy

- Built a framework making Grad-CAM heatmaps invariant to rotation, zoom, and shift via inverse-realignment and averaging, with no model retraining.
- Implemented the pipeline in TensorFlow/Keras across ResNet152, DenseNet201, and Xception on natural and histopathology datasets.
- Improved explanation faithfulness (AUC-ROC) by up to 6% using L2 and AUC-ROC metrics.

Bug Severity Classification with LLMs [\[Github\]](#) | PyTorch, CodeBERT, Hugging Face, NLP

- Fine-tuned CodeBERT with an NT-Xent contrastive objective to classify bug severity from 13K+ reports, reaching up to 98% F1.
- Boosted cross-project accuracy on unseen projects by 56%, staying robust under class imbalance that sank baselines below 50%.
- Outperformed Doc2Vec and non-contrastive baselines; used t-SNE to visualize sharper severity clusters.

Forecasting Stock Market Trends [\[Github\]](#) | Python, TensorFlow/Keras, yfinance, Tkinter

- Built a desktop app integrating four ML models (Feedforward NN, LSTM, Random Forest Classifier & Regressor) for fundamental, trading, risk, and portfolio analysis on tech stocks
- Engineered a time-series pipeline with live yfinance data, moving-average feature engineering, and RMSE/accuracy evaluation
- Packaged all models behind a tabbed Tkinter GUI for real-time predictions and visualization

Satellite Terrain Classification [\[Github\]](#) | Python, PyTorch, scikit-learn, Shiny

- Built and trained a custom CNN from scratch in PyTorch (1.2M params) achieving 92.6% accuracy across 10 terrain classes on 10,000 satellite images
- Engineered an end-to-end pipeline with data augmentation, cosine-annealing LR scheduling, and early stopping, evaluated via confusion matrix and error analysis
- Deployed the model as an interactive Shiny web app with real-time image upload and per-class confidence scoring

Education

Ontario Tech University

Graduated: December 2025

M.Sc in Computer Science (Specialization in Artificial Intelligence) (GPA: 3.92 / 4.30)

Oshawa, Ontario

- **Relevant Coursework:** Machine Learning, Applications of LLMs to Software Engineering, IT Security, Research Methods & Collaborative Design

North South University

Graduated: August 2020

B.Sc in Computer Science and Engineering (GPA: 3.67 / 4.00)

Dhaka, Bangladesh

- **Relevant Coursework:** Data Structure & Algorithm, Software Engineering, Database Management Systems, Probability & Statistics, Linear Algebra, Calculus (I-IV), Design & Analysis of Algorithms

Publications

1. **A Contrastive Learning Approach to Bug Severity Classification with Large Language Model Embeddings**
Conference on Computers, Software, and Applications Conference (COMPSAC 2025)
2. **I-Grad-CAM: Transformation Invariant Grad-CAM**
Conference on Vision and Intelligent Systems (CVIS 2024)